

BENOUALI ABDELKADER YAHIA ZAKARIA

Master's Student in AI and Cybersecurity • AI Software Engineering Intern Candidate
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Open to Relocation and Remote Roles

PROFESSIONAL SUMMARY

Master's student in AI and Cybersecurity (Ranked 3rd in CS cohort) proficient in Python, C/C++, PyTorch, and Scikit-learn. Demonstrated track record in building end-to-end AI pipelines, model quantization, and deploying cloud microservices with strict privacy and security controls. Seeking the AI Software Engineering Graduate Intern role at Intel to optimize AI frameworks and drive hardware-software co-optimization.

TECHNICAL SKILLS

- **AI and Machine Learning:** PyTorch, Scikit-learn, Model Quantization, Fine-Tuning, LLM Pipelines, Sensor & Time-Series Data Modeling
- **Languages and Core CS:** Python, C, C++, Data Structures, Algorithms, REST APIs, SQL, PostgreSQL
- **Systems and Hardware Integration:** Hardware-Software Integration, Docker Containerization, Linux/Windows Administration, Streamlit, Git/GitHub
- **Engineering and Benchmarking:** Performance Benchmarking, Parallel Computing, REST API Design, Software Debugging

EDUCATION

University Ain Temouchent Belhadj Bouchaïb, Algeria

Master of Science in Computer and Information Sciences - AI and Cybersecurity Sept 2025 - July 2027 (Expected)

Bachelor of Science in Computer Science Sept 2022 - June 2025

- **Academic Distinction:** Ranked 3rd out of cohort across L1 and L2 semesters. Core Coursework: C Programming, Data Structures, Algorithms, Computer Architecture, Operating Systems.

AI AND SOFTWARE ENGINEERING PROJECTS

Adaptive Blended Assessment (WriteLens V2) 2025 - Present (Ongoing)

github.com/benoualiabdelkader/Adaptive-Blended-Assessment- Tech Stack: Python, Node.js, Express, React, TypeScript, Docker

- Architected and built an AI-driven academic writing platform processing 480+ student submissions to generate diagnostic profiles, risk bands, and cohort summaries for 60 students.
- Built a privacy-first Python AI engine integrating Groq LLM API with strict tokenization, default-deny security controls, and provenance tracking across Render cloud deployments.
- Developed a REST API backend (Node.js/Express) serving JSON run bundles to a React TypeScript dashboard, containerized via Docker with release automation.

EdgeQuant - High-Speed LLM Quantization Toolkit

2026 (Completed)

github.com/benoualiabdelkader/EdgeQuant

Tech Stack: Python, C++, PyTorch, CUDA, ONNX

- Designed and implemented an open-source quantization toolkit compressing 7B and 13B parameter Transformer models into 4-bit and 8-bit precision for edge devices and GPUs.
- Optimized inference performance by reducing GPU VRAM footprint by 62% while preserving 98.4% of baseline FP16 model accuracy.

EcoSentinel AI - Aquatic Sensor Quality and ML Pipeline

2026 (Completed)

github.com/benoualiabdelkader/eco-sentinel-project

Tech Stack: Python, Scikit-learn, Pandas, NumPy, Streamlit, L^AT_EX

- Designed and built an environmental monitoring system processing aquatic sensor data (turbidity, dissolved oxygen) to detect water pollution in real time.
- Trained a Scikit-learn SVM classifier (RBF kernel) and deployed an interactive Streamlit simulation web app featuring real-time parameter sliders and published a formal L^AT_EX research report.

LEADERSHIP AND GLOBAL PROGRAMS

Aspire Leaders Program Participant - Aspire Institute (Harvard-affiliated)

July 2026 - Present

- Selected for a global leadership program; completing coursework and masterclasses on strategic decision-making and technical leadership.

Participant - Erasmus+ Virtual Exchange - VIRTUALLYEDU (EU-Funded)

July 2026 - Present

- Engaging in an EU-funded virtual exchange focused on digital skills, cybersecurity defense strategies, and cross-cultural technical collaboration.

CERTIFICATIONS

- **Google AI Essentials Certification:** Generative AI, Responsible AI, Data Ethics
- **Google IT Support Professional Certificate:** Linux System Administration, Computer Networking, Hardware Troubleshooting
- **Algerian-American Summer University 2025:** Advanced Computing Track (AI, Machine Learning, Cybersecurity and Ethics)

LANGUAGES

Arabic: Native • English: B2 Level (Intermediate / Working Proficiency) • French: B1 Level (Working Proficiency)